

AKS Observability Platform - PowerShell Runbook

Command-by-command implementation guide with validation screenshots



CLI path: PowerShell + Docker + Azure CLI + kubectl + Git

Document	Prepared for	Date	Repository
AKS Observability Platform - PowerShell Runbook	Rizwan Siddiqui	30 July 2026	RizwanSid7/azure-monitor-aks-observability-capstone

Use of this document

This PDF is written so another learner or engineer can understand why this setup exists, how to implement it, how to validate it, and how to present it in a demo.

1. What This Runbook Builds

Section goal
Reproduce the project from local app to monitored AKS deployment.

This runbook contains the command-line implementation for the AKS Observability Capstone project. It is written as a reliable command reference with explanation, validation commands, troubleshooting fixes, and screenshots.

Layer	Implementation
Local application	Flask app with /, /health, /error, /cpu, and /crash endpoints.
Containerization	Dockerfile and Docker Desktop local test.
Registry	Azure Container Registry used to store images.
Orchestration	Azure Kubernetes Service with deployment, service, namespace, and ingress.
Ingress	Application Gateway Ingress Controller routes traffic from App Gateway to AKS service.
Observability	Container Insights, AMA logs, Log Analytics Workspace, KQL, kubectl logs, crash/restart test.
CI/CD	Azure DevOps pipeline builds and pushes image to ACR through self-hosted agent.

2. Prerequisites and One-Time Setup

Section goal

Install tools and configure Git identity before starting.

Tool	Purpose
Git	Version control, GitHub push, commit history.
Docker Desktop	Build and run the local container and support Azure DevOps self-hosted agent build task.
Azure CLI	Create and manage Azure resources.
kubectl	Apply Kubernetes manifests and validate pods/services/ingress.
Azure DevOps PAT	Registers the local machine as a self-hosted agent.
PowerShell	Main shell used for Azure CLI and Windows commands.

```
git config --global user.name "Rizwan Siddiqui";
git config --global user.email "lt;your-email>";

# Verify Docker Desktop is running
docker version
docker ps

# Verify Azure CLI
az version

# Login if required
az login --use-device-code
az account show --output table
```

```

MINGW64/c/Users/Admin/Documents/azure-monitor-aks-observability-capstone
On branch master
No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    Dockerfile
    app/
    azure-pipelines.yml
    k8s/
    monitoring/
    presentation/
    screenshots/

nothing added to commit but untracked files present (use "git add" to track)
warning: in the working copy of 'Dockerfile', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'app/app.py', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'app/requirements.txt', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'azure-pipelines.yml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'k8s/deployment.yaml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'k8s/ingress.yaml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'k8s/service.yaml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'monitoring/alert-rules.md', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'monitoring/kql-queries.md', LF will be replaced by CRLF the next time Git touches it
Author identity unknown

*** Please tell me who you are.

Run

  git config --global user.email
  git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: unable to auto-detect email address (got 'Admin@Rizwan.(none)')

admin@Rizwan MINGW64 ~/Documents/azure-monitor-aks-observability-capstone (master)
$ |

```

Initial project structure / Git identity lesson.

```

Select a [redacted] and tenant (Type a number or Enter for no changes): 1

Tenant [redacted]

[Announcements]
With the new Azure CLI login experience, you can see [redacted] and its configuration at https://go.microsoft.com/fwlink/?linkid=2271236

If you encounter any problem, please open an issue at https://aka.ms/azclibug

[Warning] The login output has been updated. Please be aware that it no longer displays the full list of available [redacted] by default.

PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone>

```

Azure account verification screenshot.

3. Create Local Project Structure

Section goal

Create folders and files for app, Kubernetes, monitoring, docs, and screenshots.

```
cd $env:USERPROFILE\Documents
mkdir azure-monitor-aks-observability-capstone
cd azure-monitor-aks-observability-capstone

git init
mkdir app,k8s,monitoring,scripts,screenshots,presentation,docs
New-Item app\app.py, app\requirements.txt, Dockerfile -ItemType File
New-Item k8s\namespace.yaml, k8s\deployment.yaml, k8s\service.yaml, k8s\ingress.yaml -ItemType File
New-Item monitoring\kql-queries.md, monitoring\alert-rules.md, azure-pipelines.yml, README.md -ItemType File

git add .
git commit -m "Initial AKS observability capstone project structure"
git branch -M main
```

```
MINGW64/c:/Users/Admin/Documents/azure-monitor-aks-observability-capstone
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    Dockerfile
    app/
    azure-pipelines.yml
    k8s/
    monitoring/
    presentation/
    screenshots/

nothing added to commit but untracked files present (use "git add" to track)
warning: in the working copy of 'Dockerfile', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'app/app.py', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'app/requirements.txt', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'azure-pipelines.yml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'k8s/deployment.yaml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'k8s/ingress.yaml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'k8s/service.yaml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'monitoring/alert-rules.md', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'monitoring/kql-queries.md', LF will be replaced by CRLF the next time Git touches it
Author identity unknown

*** Please tell me who you are.

Run

  git config --global user.email
  git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: unable to auto-detect email address (got 'Admin@Rizwan.(none)')

Admin@Rizwan MINGW64 ~/Documents/azure-monitor-aks-observability-capstone (master)
$ |
```

Folder structure after creating initial project files.

4. Application Code and Dockerfile

Section goal

Create a simple app that generates normal logs, error logs, CPU load, and crash behavior.

The app exposes the following endpoints: / for home page, /health for health validation, /error to generate an application error log, /cpu to generate load, and /crash to trigger restart behavior.

```
# app/requirements.txt
flask==3.0.3

# Dockerfile
FROM python:3.11-slim
WORKDIR /app
COPY app/requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY app/ .
EXPOSE 5000
CMD ["python", "app.py"]
```

```

MINGW64/c/Users/Admin/Documents/azure-monitor-aks-observability-capstone
CONTAINER ID    IMAGE    COMMAND    CREATED    STATUS    PORTS    NAMES
Admin@Rizwan MINGW64 ~/Documents/azure-monitor-aks-observability-capstone (main)
$ docker build -t observability-app:local .
[+] Building 22.4s (11/11) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 208B
=> [internal] load metadata for docker.io/library/python:3.11-slim
=> [auth] library/python:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load build context
=> => transferring context: 1.42kB
=> [1/5] FROM docker.io/library/python:3.11-slim@sha256:db3ff2e1800a8581e2c48a27c3995339d47bdf046da21c7627accd3d51053a93
=> => resolve docker.io/library/python:3.11-slim@sha256:db3ff2e1800a8581e2c48a27c3995339d47bdf046da21c7627accd3d51053a93
=> => sha256:c89b9f64c028c19ba92e195d7589d914b9cd1fc69f3c3dfef931e93876ac2064 248B / 248B
=> => sha256:6b265b8eae4a26a263eb55546aa430c1eef37032a7cd86393d98c6b779f28319 14.42MB / 14.42MB
=> => sha256:062e450697faa5f02a3a74eba9864ee4d79bc9cfbd65769fc6cdf2c05c6a053 29.78MB / 29.78MB
=> => sha256:9775d166087aba0afe5ba9a88859c04c08090e82ae218742d09fec9119e0335d 1.29MB / 1.29MB
=> => extracting sha256:062e450697faa5f02a3a74eba9864ee4d79bc9cfbd65769fc6cdf2c05c6a053
=> => extracting sha256:9775d166087aba0afe5ba9a88859c04c08090e82ae218742d09fec9119e0335d
=> => extracting sha256:6b265b8eae4a26a263eb55546aa430c1eef37032a7cd86393d98c6b779f28319
=> => extracting sha256:c89b9f64c028c19ba92e195d7589d914b9cd1fc69f3c3dfef931e93876ac2064
=> [2/5] WORKDIR /app
=> [3/5] COPY app/requirements.txt .
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt
=> [5/5] COPY app/ .
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:d0c3633e6909ac908a31843a346e9c3afe61b7971cb4a6c34fec8ddce47dc979
=> => exporting config sha256:8c0943f12959d073ed4dc3df2283b27d7e71e4f38376bee8fd8ef3db0ffdb49f
=> => exporting attestation manifest sha256:89609fdf5e713adca8839f1a9933c9b551509797ebd6a9b82cb97d27700d8116
=> => exporting manifest list sha256:6c6872e34d9e1e067a8af140a81b8d53950f962542dac6dd79fe66c7b5b32465
=> => naming to docker.io/library/observability-app:local
=> => unpacking to docker.io/library/observability-app:local
View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/jbde4vs96k1i2jbjtew0iy7h5
Admin@Rizwan MINGW64 ~/Documents/azure-monitor-aks-observability-capstone (main)

```

Docker image build successful locally.

5. Local Docker Build and Test

Section goal

Validate the app before deploying to Azure.

```
docker build -t observability-app:local .

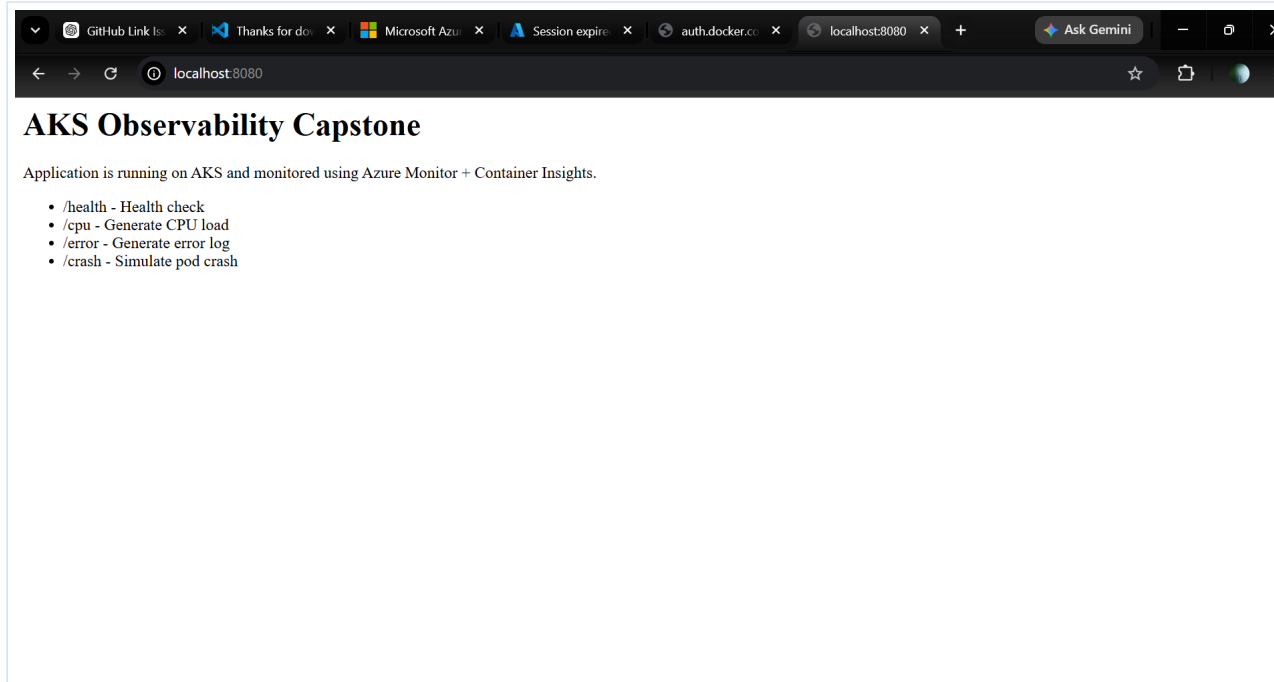
docker run -d --name observability-app-local -p 8080:5000 observability-app:local

docker ps

# Browser test
# http://localhost:8080

# Endpoint tests
curl http://localhost:8080/
curl http://localhost:8080/health
curl http://localhost:8080/error
curl http://localhost:8080/cpu

# Logs
docker logs observability-app-local --tail 50
```



Local browser validation.

```
Admin@Rizwan MINGW64 ~/Documents/azure-monitor-aks-observability-capstone (main)
$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS                               NAMES
56e980cebdca   observability-app:local             "python app.py"         57 seconds ago Up 56 seconds 0.0.0.0:8080->5000/tcp, [::]:8080->5000/tcp   observability-app-local

Admin@Rizwan MINGW64 ~/Documents/azure-monitor-aks-observability-capstone (main)
$
```

Docker container running.

```
Admin@Rizwan MINGW64 ~/Documents/azure-monitor-aks-observability-capstone (main)
$ curl http://localhost:8080/
curl http://localhost:8080/health
curl http://localhost:8080/error
curl http://localhost:8080/cpu

<h1>AKS Observability Capstone</h1>
<p>Application is running on AKS and monitored using Azure Monitor + Container Insights.</p>
<ul>
  <li>/health - Health check</li>
  <li>/cpu - Generate CPU load</li>
  <li>/error - Generate error log</li>
  <li>/crash - Simulate pod crash</li>
</ul>
{"service":"aks-observability-app","status":"healthy"}
Error log generated. Check Container Logs in Log Analytics.CPU load generated for monitoring test
Admin@Rizwan MINGW64 ~/Documents/azure-monitor-aks-observability-capstone (main)
$
```

Local endpoint tests.

```
Admin@Rizwan MINGW64 ~/Documents/azure-monitor-aks-observability-capstone (main)
$ docker logs observability-app-local --tail 50
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on
Press CTRL+C to quit
- - [28/Jul/2026 17:36:40] "GET / HTTP/1.1" 200 -
- - [28/Jul/2026 17:36:41] "GET /favicon.ico HTTP/1.1" 404 -
- - [28/Jul/2026 17:38:05] "GET / HTTP/1.1" 200 -
- - [28/Jul/2026 17:38:05] "GET /health HTTP/1.1" 200 -
[2026-07-28 17:38:05,901] ERROR in app: Sample ERROR log generated for Azure Monitor KQL testing
- - [28/Jul/2026 17:38:05] "GET /error HTTP/1.1" 200 -
[2026-07-28 17:38:06,400] WARNING in app: CPU load endpoint triggered
- - [28/Jul/2026 17:38:16] "GET /cpu HTTP/1.1" 200 -

Admin@Rizwan MINGW64 ~/Documents/azure-monitor-aks-observability-capstone (main)
$
```

Local Docker logs show /error and /cpu events.

6. Push Code to GitHub

Section goal

Version control and remote repository proof.

```
git status
git add .
git commit -m "Tested observability app locally with Docker"

# Create GitHub repo first, then add remote
git remote add origin https://github.com/RizwanSid7/azure-monitor-aks-observability-capstone.git
git push -u origin main
```

```
Admin@Rizwan MINGW64 ~/Documents/azure-monitor-aks-observability-capstone (main)
$ cd ~/Documents/azure-monitor-aks-observability-capstone

git status
git log --oneline --max-count=3
On branch main
nothing to commit, working tree clean
d6376ac (HEAD -> main) Tested observability app locally with Docker
879d383 Initial AKS observability capstone project structure

Admin@Rizwan MINGW64 ~/Documents/azure-monitor-aks-observability-capstone (main)
$ git remote add origin https://github.com/RizwanSid7/azure-monitor-aks-observability-capstone.git
git push -u origin main
info: please complete authentication in your browser...
Enumerating objects: 25, done.
Counting objects: 100% (25/25), done.
Delta compression using up to 8 threads
Compressing objects: 100% (22/22), done.
Writing objects: 100% (25/25), 489.87 KiB | 15.31 MiB/s, done.
Total 25 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/RizwanSid7/azure-monitor-aks-observability-capstone.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

Admin@Rizwan MINGW64 ~/Documents/azure-monitor-aks-observability-capstone (main)
$
```

GitHub repository created and pushed.

7. Define Azure Variables

Section goal

Use variables so commands stay clean and repeatable.

```
$RG="rg-aks-observability-capstone";  
$LOC="centralindia";  
$ACR="acr-rizwanobs268";  
$AKS="aks-observability-capstone";  
$LAW="law-aks-observability";  
$APPGW="appgw-observability";  
$APP_IMAGE="observability-app";  
$APP_URL="http://20.207.126.15";
```

Why variables help

They reduce typing mistakes and make it easy to reuse the same command set in demos or documentation.

8. Create Resource Group

Section goal

All lab resources are grouped for clean management and deletion.

```
az group create --name $RG --location $LOC
az group show --name $RG --output table
```

```
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> $RG="rg-aks-observability-capstone"
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> $LOC="centralindia"
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> $ACR="acr rizwanobs268"
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> $AKS="aks-observability-capstone"
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> $LAW="law-aks-observability"
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> $APPGW="appgw-observability"
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> $APP_IMAGE="observability-app"
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> az group create --name $RG --location $LOC
{
  "location": "centralindia",
  "managedBy": null,
  "name": "rg-aks-observability-capstone",
  "properties": {
    "provisioningState": "Succeeded"
  },
  "tags": null,
  "type": "Microsoft.Resources/resourceGroups"
}
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone>
```

Resource group created successfully.

9. Create ACR and Build/Push Image

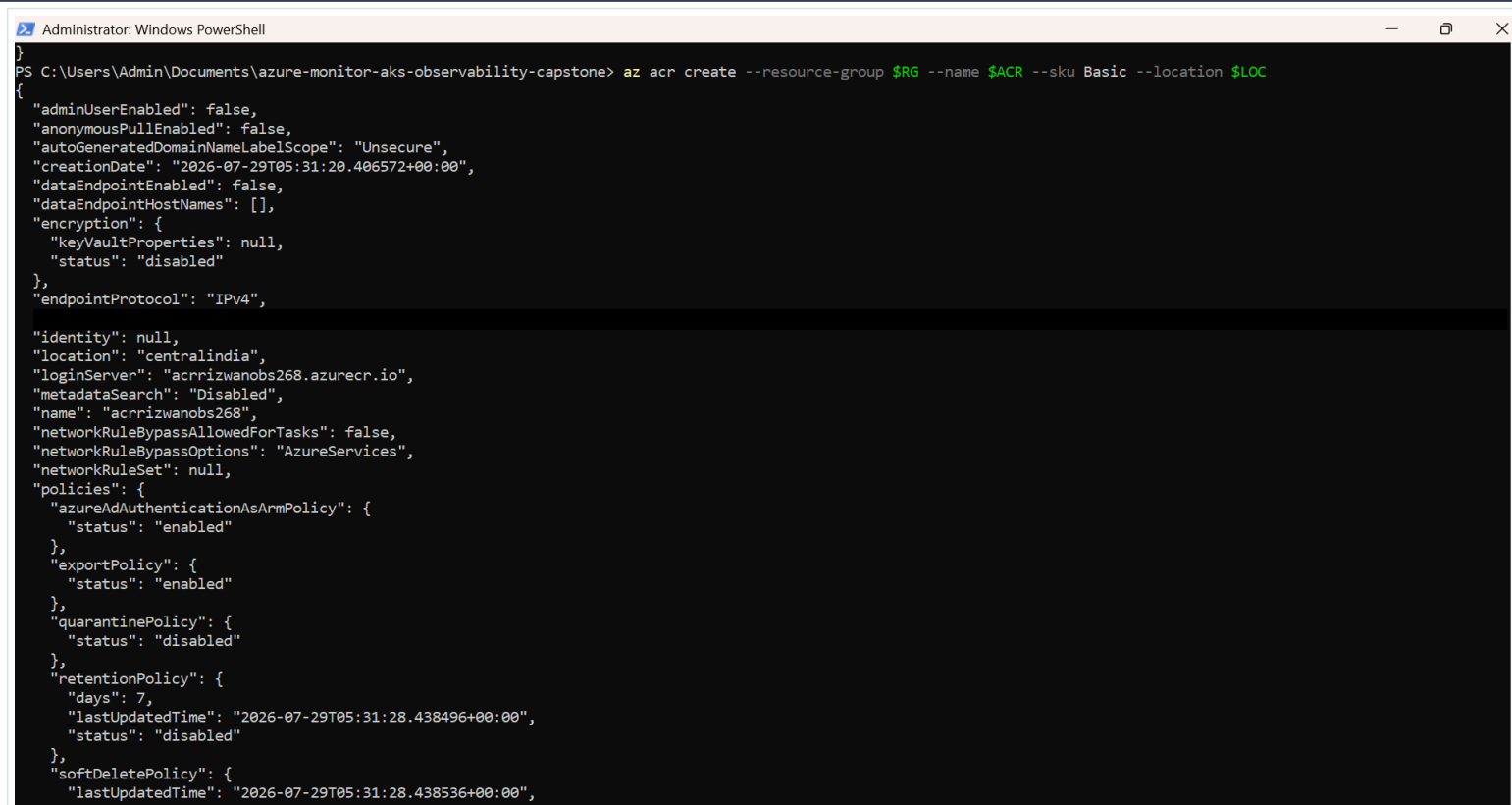
Section goal

Create registry and push observability-app:latest.

```
az acr create --resource-group $RG --name $ACR --sku Basic --location $LOC

az acr build --registry $ACR --image &quot;$APP_IMAGE`:latest&quot; .

az acr repository list --name $ACR --output table
az acr repository show-tags --name $ACR --repository $APP_IMAGE --output table
```



```
Administrator: Windows PowerShell
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> az acr create --resource-group $RG --name $ACR --sku Basic --location $LOC
{
  "adminUserEnabled": false,
  "anonymousPullEnabled": false,
  "autoGeneratedDomainNameLabelScope": "Unsecure",
  "creationDate": "2026-07-29T05:31:20.406572+00:00",
  "dataEndpointEnabled": false,
  "dataEndpointHostNames": [],
  "encryption": {
    "keyVaultProperties": null,
    "status": "disabled"
  },
  "endpointProtocol": "IPv4",
  "identity": null,
  "location": "centralindia",
  "loginServer": "acrriizwanobs268.azurecr.io",
  "metadataSearch": "Disabled",
  "name": "acrriizwanobs268",
  "networkRuleBypassAllowedForTasks": false,
  "networkRuleBypassOptions": "AzureServices",
  "networkRuleSet": null,
  "policies": {
    "azureAdAuthenticationAsArmPolicy": {
      "status": "enabled"
    },
    "exportPolicy": {
      "status": "enabled"
    },
    "quarantinePolicy": {
      "status": "disabled"
    },
    "retentionPolicy": {
      "days": 7,
      "lastUpdatedTime": "2026-07-29T05:31:28.438496+00:00",
      "status": "disabled"
    },
    "softDeletePolicy": {
      "lastUpdatedTime": "2026-07-29T05:31:28.438536+00:00",

```

ACR created.

```

Administrator: Windows PowerShell
memory: "512Mi"
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> az acr build --registry $ACR --image "$APP_IMAGE":latest" .
Packing source code into tar to upload...
Excluding '.git' based on default ignore rules
Uploading archived source code from
Sending context (574.769 KiB) to registry: acrrizwanobs268...
Queued a build with ID: cu1
Waiting for an agent...
2026/07/29 05:34:03 Downloading source code...
2026/07/29 05:34:04 Finished downloading source code
2026/07/29 05:34:04 Using acb_vol_e6409529-99f7-4707-8f5f-d52bd9d865d6 as the home volume
2026/07/29 05:34:04 Setting up Docker configuration...
2026/07/29 05:34:05 Successfully set up Docker configuration
2026/07/29 05:34:05 Logging in to registry: acrrizwanobs268.azurecr.io
2026/07/29 05:34:10 Successfully logged into acrrizwanobs268.azurecr.io
2026/07/29 05:34:10 Executing step ID: build. Timeout(sec): 28800, Working directory: '', Network: ''
2026/07/29 05:34:10 Scanning for dependencies...
2026/07/29 05:34:11 Successfully scanned dependencies
2026/07/29 05:34:11 Launching container with name: build
Sending build context to Docker daemon 665.6kB
Step 1/7 : FROM python:3.11-slim
3.11-slim: Pulling from library/python
062e450697fa: Pulling fs layer
9775d166087a: Pulling fs layer
6b265b8eae4a: Pulling fs layer
c89b9f64c028: Pulling fs layer
c89b9f64c028: Waiting
9775d166087a: Verifying Checksum
9775d166087a: Download complete
6b265b8eae4a: Verifying Checksum
6b265b8eae4a: Download complete
062e450697fa: Verifying Checksum
062e450697fa: Download complete
c89b9f64c028: Verifying Checksum
c89b9f64c028: Download complete
062e450697fa: Pull complete
9775d166087a: Pull complete
6b265b8eae4a: Pull complete

```

ACR build pushed image to registry.

```

Run ID: cu1 was successful after 34s
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> az acr repository list --name $ACR --output table
Result
-----
observability-app
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> az acr repository show-tags --name $ACR --repository $APP_IMAGE --output table
Result
-----
latest
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> _

```

ACR repository tags show latest.

10. Create Log Analytics Workspace

Section goal

Workspace stores logs/metrics for KQL and Container Insights.

```
az monitor log-analytics workspace create `
  --resource-group $RG `
  --workspace-name $LAW `
  --location $LOC

$LAW_ID = az monitor log-analytics workspace show `
  --resource-group $RG `
  --workspace-name $LAW `
  --query id -o tsv

$WORKSPACE_ID = az monitor log-analytics workspace show `
  --resource-group $RG `
  --workspace-name $LAW `
  --query customerId -o tsv
```

```
latest
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> az monitor log-analytics workspace create --resource-group $RG --workspace-name $LAW --location $LOC
{
  "createdDate": "2026-07-29T05:49:08.9185003Z",
  "etag": "0x86c3b1b1b1b1b1b1",
  "features": {
    "enableLogAccessUsingOnlyResourcePermissions": true,
    "legacy": 0,
    "searchVersion": 1
  },
  "location": "centralindia",
  "modifiedDate": "2026-07-29T05:49:22.3011708Z",
  "name": "law-aks-observability",
  "provisioningState": "Succeeded",
  "publicNetworkAccessForIngestion": "Enabled",
  "publicNetworkAccessForQuery": "Enabled",
  "resourceGroup": "rg-aks-observability-capstone",
  "retentionInDays": 30,
  "sku": {
    "lastSkuUpdate": "2026-07-29T05:49:08.9185003Z",
    "name": "PerGB2018"
  },
  "type": "Microsoft.OperationalInsights/workspaces",
  "workspaceCapping": {
    "dailyQuotaGb": -1.0,
    "dataIngestionStatus": "RespectQuota",
    "quotaNextResetTime": "2026-07-29T10:00:00Z"
  }
}
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone>
```

Log Analytics workspace created.

11. Create AKS with Monitoring and AGIC

Section goal

Provision AKS with Container Insights and Application Gateway Ingress Controller.

```
az aks create `
  --resource-group $RG `
  --name $AKS `
  --location $LOC `
  --node-count 1 `
  --node-vm-size Standard_B2s `
  --network-plugin azure `
  --enable-managed-identity `
  --enable-addons monitoring,ingress-appgw `
  --workspace-resource-id $LAW_ID `
  --appgw-name $APPGW `
  --appgw-subnet-cidr "10.225.0.0/16"; `
  --generate-ssh-keys

az aks get-credentials --resource-group $RG --name $AKS --overwrite-existing
kubectl get nodes

az aks show --resource-group $RG --name $AKS --query "addonProfiles" --output json
```

```
"windowsProfile": {
  "adminPassword": null,
  "adminUsername": "azureuser",
  "enableCsiProxy": true,
  "gmsaProfile": null,
  "licenseType": null
},
"workloadAutoScalerProfile": {
  "keda": null,
  "verticalPodAutoscaler": null
}
}
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> az aks get-credentials --resource-group $RG --name $AKS --overwrite-existing
Merged "aks-observability-capstone" as current context in C:\Users\Admin\.kube\config
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> kubectl get nodes
NAME                                STATUS    ROLES    AGE     VERSION
aks-nodepool1-30392603-vmss000000  Ready    <none>    4h10m  v1.35.6
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone>
```

AKS node ready.

```

Merged "aks-observability-capstone" as current context in C:\Users\Admin\.kube\config
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> kubectl get nodes
NAME                                STATUS    ROLES    AGE   VERSION
aks-nodepool1-30392603-vmss000000  Ready    <none>   4h10m v1.35.6
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> az aks show --resource-group $RG --name $AKS --query "addonProfiles" --output json
{
  "extensionManager": {
    "config": null,
    "enabled": true,
    "identity": null
  },
  "ingressApplicationGateway": {
    "config": {
      "applicationGatewayName": "appgw-observability",
      "subnetCIDR": "10.225.0.0/16",
      "enabled": true,
      "identity": {
        "type": "UserAssigned",
        "userAssignedIdentities": {
          "microsoft.ManagedIdentity/userAssignedIdentities/ingressapplicationgateway-aks-observability-capstone": {}
        }
      },
      "omsagent": {
        "config": {
          "workspaces/law-aks-observability",
          "useAADAuth": "true"
        },
        "enabled": true,
        "identity": null
      }
    }
  }
}
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone>

```

AKS add-ons verified: AGIC and omsagent enabled.

```
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> az aks show --resource-group $RG --name $AKS --query "addonProfiles" --output json
{
  "extensionManager": {
    "config": null,
    "enabled": true,
    "identity": null
  },
  "ingressApplicationGateway": {
    "config": {
      "applicationGatewayName": "appgw-observability",
      "effectiveApplicationGatewayId": "/subscriptions/{REDACTED RESOURCE ID}
alindia/providers/Microsoft.Network/applicationGateways, ...
      "subnetCIDR": REDACTED
    },
    "enabled": true,
    "identity": {
      "clientId": "7! REDACTED GUIDS
      "objectId": "d
      "resourceId": 'REDACTED RESOURCE ID
icrosoft.ManagedIdenti
    },
    "omsagent": {
      "config": {
        "logAnalyticsWorkspaceResourceId": "/sub REDACTED WORKSPACE RESOURCE ID
REDACTED
        "useAADAuth": "true"
      },
      "enabled": true,
      "identity": null
    }
  }
}
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> _
```

Later add-ons verification with resource IDs redacted.

12. Kubernetes Manifest Files

Section goal

Deploy namespace, deployment, service, and ingress.

```
# namespace.yaml
apiVersion: v1
kind: Namespace
metadata:
  name: observability-demo

# deployment.yaml uses image:
# acrrizwanobs268.azurecr.io/observability-app:latest

# service.yaml exposes port 80 to targetPort 5000 as ClusterIP.
# ingress.yaml maps / to observability-service:80 using AGIC annotation:
# kubernetes.io/ingress.class: azure/application-gateway
```

```
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> kubectl apply -f .\k8s\namespace.yaml
namespace/observability-demo created
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> kubectl apply -f .\k8s\deployment.yaml
deployment.apps/observability-app created
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> kubectl apply -f .\k8s\service.yaml
service/observability-service created
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> kubectl apply -f .\k8s\ingress.yaml
```

Manifest files applied.

13. Apply Kubernetes Manifests and Verify Resources

Section goal

Deploy application to AKS.

```
kubectl apply -f .\k8s\namespace.yaml
kubectl apply -f .\k8s\deployment.yaml
kubectl apply -f .\k8s\service.yaml
kubectl apply -f .\k8s\ingress.yaml

kubectl rollout status deployment/observability-app -n observability-demo
kubectl get all -n observability-demo
kubectl get ingress -n observability-demo
```

```
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> kubectl apply -f .\k8s\namespace.yaml
namespace/observability-demo unchanged
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> kubectl apply -f .\k8s\deployment.yaml
deployment.apps/observability-app configured
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> kubectl apply -f .\k8s\service.yaml
service/observability-service unchanged
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> kubectl apply -f .\k8s\ingress.yaml
ingress.networking.k8s.io/observability-ingress unchanged
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone>
```

Kubernetes files applied/unchanged after re-apply.

```

    cpu: "100m"
    memory: "128Mi"
  limits:
    cpu: "500m"
    memory: "512Mi"
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> kubectl rollout status deployment/observability-app -n observability-demo
deployment "observability-app" successfully rolled out
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> kubectl get all -n observability-demo
NAME                                READY   STATUS    RESTARTS   AGE
pod/observability-app-654cccc487-pgch9  1/1     Running   0           82m
pod/observability-app-654cccc487-sqshp  1/1     Running   0           82m

NAME                                TYPE               CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
service/observability-service        ClusterIP           <redacted>        <none>            80/TCP           90m

NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/observability-app    2/2     2             2           90m

NAME                                DESIRED   CURRENT   READY   AGE
replicaset.apps/observability-app-5c4d8dc8c  0         0         0       90m
replicaset.apps/observability-app-654cccc487  2         2         2       82m
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> kubectl get ingress -n observability-demo
NAME            CLASS    HOSTS      ADDRESS      PORTS    AGE
observability-ingress  <none>  *          <redacted>    80       86m
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone>

```

Pods, deployment, service, and ingress verification.

```

Ingress.networking.k8s.io/observability-ingress unchanged
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> kubectl get ingress -n observability-demo
NAME            CLASS    HOSTS      ADDRESS      PORTS    AGE
observability-ingress  <none>  *          REDACTED IP  5        80       6h32m
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone>

```

Ingress created with Application Gateway address.

14. Test Application Through Application Gateway

Section goal

Use endpoint paths to generate monitoring signals.

```
curl.exe http://20.207.126.15/health
curl.exe http://20.207.126.15/error
curl.exe http://20.207.126.15/cpu
```

```
kubectl logs -n observability-demo deployment/observability-app --tail=80
```

```
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> curl.exe REDACTED URL alth
{"service":"aks-observability-app","status":"healthy"}
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> curl.exe http://20.207.126.15/error
Error log generated. Check Container Logs in Log Analytics. REDACTED URL
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> curl.exe http://20.207.126.15/cpu
CPU load generated for monitoring test REDACTED URL
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone>
```

Health, error, and CPU endpoints tested through Application Gateway.

```

CPU load generated for monitoring test
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> kubectl logs -n observability-demo deployment/observability-app --tail=80
IP      pods, using pod/observability-app-654cccc487-pgch9
.5 - - [29/Jul/2026 16:42:21] "GET / HTTP/1.1" 200 -
.4 - - [29/Jul/2026 16:42:23] "GET / HTTP/1.1" 200 -
.5 - - [29/Jul/2026 16:42:51] "GET / HTTP/1.1" 200 -
.4 - - [29/Jul/2026 16:42:53] "GET / HTTP/1.1" 200 -
.5 - - [29/Jul/2026 16:43:21] "GET / HTTP/1.1" 200 -
.4 - - [29/Jul/2026 16:43:23] "GET / HTTP/1.1" 200 -
.5 - - [29/Jul/2026 16:43:51] "GET / HTTP/1.1" 200 -
.4 - - [29/Jul/2026 16:43:53] "GET / HTTP/1.1" 200 -
.5 - - [29/Jul/2026 16:44:21] "GET / HTTP/1.1" 200 -
.4 - - [29/Jul/2026 16:44:23] "GET / HTTP/1.1" 200 -
.5 - - [29/Jul/2026 16:44:51] "GET / HTTP/1.1" 200 -
.4 - - [29/Jul/2026 16:44:53] "GET / HTTP/1.1" 200 -
.5 - - [29/Jul/2026 16:45:21] "GET / HTTP/1.1" 200 -
.4 - - [29/Jul/2026 16:45:23] "GET / HTTP/1.1" 200 -
.5 - - [29/Jul/2026 16:45:51] "GET / HTTP/1.1" 200 -
.4 - - [29/Jul/2026 16:45:53] "GET / HTTP/1.1" 200 -
.5 - - [29/Jul/2026 16:46:21] "GET / HTTP/1.1" 200 -
.4 - - [29/Jul/2026 16:46:23] "GET / HTTP/1.1" 200 -
.5 - - [29/Jul/2026 16:46:51] "GET / HTTP/1.1" 200 -
.4 - - [29/Jul/2026 16:46:53] "GET / HTTP/1.1" 200 -
.5 - - [29/Jul/2026 16:47:21] "GET / HTTP/1.1" 200 -
.4 - - [29/Jul/2026 16:47:23] "GET / HTTP/1.1" 200 -
.5 - - [29/Jul/2026 16:47:51] "GET / HTTP/1.1" 200 -
.4 - - [29/Jul/2026 16:47:53] "GET / HTTP/1.1" 200 -
.5 - - [29/Jul/2026 16:48:21] "GET / HTTP/1.1" 200 -
.4 - - [29/Jul/2026 16:48:23] "GET / HTTP/1.1" 200 -
10.229.0.5 - - [29/Jul/2026 16:48:51] "GET / HTTP/1.1" 200 -

```

Kubernetes application logs show requests and error/warning messages.

15. Troubleshoot Ingress and Image Issues

Section goal

Record failures because they strengthen the project.

Issue	Observed symptom	Fix
InvalidImageName	Pod stuck 0/1 InvalidImageName.	Correct image path to acrrizwanobs268.azurecr.io/observability-app:latest and re-apply deployment.
App Gateway timeout	Browser timeout or curl cannot connect.	Verify ingress annotation, AGIC pod, backend health, and wait for reconciliation.
No KQL data	Log Analytics query returns blank.	Verify monitoring add-on and ama/oms pods; widen time range; use kubectl logs as immediate proof.

```
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> kubectl get pods -n observability-demo -o wide
NAME                                READY   STATUS    RESTARTS   AGE   IP              NODE                                NOMINATED NODE   READINESS GATES
observability-app-5c4d8dc8c-gt4h4   0/1     InvalidImageName  0       9m7s   REDACTED IP     aks-nodepool11-30392603-vmss000000 <none>           <none>
observability-app-654cccc487-pgch9  1/1     Running      0       6h35m   REDACTED IP     aks-nodepool11-30392603-vmss000000 <none>           <none>
observability-app-654cccc487-sqshp  1/1     Running      0       6h35m   REDACTED IP     aks-nodepool11-30392603-vmss000000 <none>           <none>
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone>
```

InvalidImageName pod and running pods visible during troubleshooting.

16. Log Analytics and KQL Commands

Section goal

Use CLI KQL as part of monitoring verification.

```
$WORKSPACE_ID = az monitor log-analytics workspace show `
  --resource-group $RG `
  --workspace-name $LAW `
  --query customerId `
  -o tsv

# Broad table check
az monitor log-analytics query --workspace $WORKSPACE_ID `
  --analytics-query &quot;search * | where TimeGenerated > ago(24h) | summarize Count=count() by $table&quot; `
  --timespan P1D --output table

# Container logs
az monitor log-analytics query --workspace $WORKSPACE_ID `
  --analytics-query &quot;ContainerLogV2 | where TimeGenerated > ago(24h) | project TimeGenerated,
    PodNamespace, PodName, LogMessage | order by TimeGenerated desc | take 20&quot; `
  --timespan P1D --output table
```

```
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> $WORKSPACE_ID = az monitor log-analytics workspace show `
>> --resource-group $RG `
>> --workspace-name $LAW `
>> --query customerId `
>> -o tsv
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone>
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> $WORKSPACE_ID
REDACTED WORKSPACE ID
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone>
```

Workspace customer ID was retrieved. Sensitive ID redacted.

```
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> curl.exe $APP_URL/health
{"service":"aks-observability-app","status":"healthy"}
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> curl.exe $APP_URL/error
Error log generated. Check Container Logs in Log Analytics.
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> curl.exe $APP_URL/cpu
CPU load generated for monitoring test
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone>
```

Traffic generated for monitoring.

```
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> az monitor log-analytics query `
>> --workspace $WORKSPACE_ID `
>> --analytics-query "ContainerLogV2 | where TimeGenerated > ago(6h) | take 10" `
>> --timespan PT6H `
>> --output table
Preview version of extension is disabled by default for extension installation, enabled for modules without stable versions.
Please run 'az config set extension.dynamic_install_allow_preview=true or false' to config it specifically.
The command requires the extension log-analytics. Do you want to install it now? The command will continue to run after the extension is installed. (Y/n): y
Run 'az config set extension.use_dynamic_install=yes_without_prompt' to allow installing extensions without prompt.
No stable version of 'log-analytics' to install. Preview versions allowed.
```

ContainerLogV2 query check.

```
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> az monitor log-analytics query --workspace $WORKSPACE_ID --analytics-query "ContainerLogV2 | where TimeGenerated > ago(24h) | project TimeGenerated, PodNamespace, PodName, LogMessage | order by TimeGenerated desc | take 20" --timespan P1D --output table

PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> kubectl get pods -n observability-demo
NAME                                READY   STATUS              RESTARTS   AGE
observability-app-5c4d8dc8c-gt4h4   0/1     InvalidImageName    0           16h
observability-app-654cccc487-pgch9  1/1     Running             0           22h
observability-app-654cccc487-sqshp  1/1     Running             0           22h
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> _
```

KQL container logs query result/validation.

The screenshot displays the Microsoft Azure portal interface for the 'law-aks-observability' Log Analytics workspace. The left-hand navigation pane shows various options, with 'Logs' currently selected. The main content area shows a query editor with the following KQL query:

```
1 Heartbeat
2 | where TimeGenerated > ago(24h)
3 | take 20
```

The query is executed, and the results section displays a message indicating that no results were found for the specified time range. The message includes a link to 'select another time range' and a link to 'Review the query details for more information'.

Heartbeat query from portal - no data case documented.

17. Crash and Restart Demo

Section goal

Demonstrate Kubernetes self-healing.

```
kubectl get pods -n observability-demo

curl.exe http://20.207.126.15/crash

Start-Sleep -Seconds 30
kubectl get pods -n observability-demo
kubectl get deployment observability-app -n observability-demo
```

```
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> kubectl get pods -n observability-demo
NAME                                READY   STATUS             RESTARTS   AGE
observability-app-5c4d8dc8c-gt4h4   0/1     InvalidImageName    0           16h
observability-app-654cccc487-pgch9  1/1     Running             0           22h
observability-app-654cccc487-sqshp  1/1     Running             0           22h
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone>
```

Pods before crash test.

```
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> curl.exe $APP_URL/crash
curl: (28) Failed to connect to REDACTED IP 5:80 after 21050 ms: Could not connect to server
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> kubectl get pods -n observability-demo
NAME                                READY   STATUS             RESTARTS   AGE
observability-app-5c4d8dc8c-gt4h4   0/1     InvalidImageName    0           16h
observability-app-654cccc487-pgch9  1/1     Running             0           22h
observability-app-654cccc487-sqshp  1/1     Running             0           22h
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> kubectl get deployment observability-app -n observability-demo
NAME            READY   UP-TO-DATE   AVAILABLE   AGE
observability-app  2/2     1             2           22h
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone>
```

Pod restart visible after crash test.

```
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> kubectl get pods -n observability-demo
```

NAME	READY	STATUS	RESTARTS	AGE
observability-app-5c4d8dc8c-gt4h4	0/1	InvalidImageName	0	16h
observability-app-654cccc487-pgch9	1/1	Running	0	23h
observability-app-654cccc487-sqshp	1/1	Running	0	23h

Second pods-before-crash screenshot.

```
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> kubectl get pods -n observability-demo
```

NAME	READY	STATUS	RESTARTS	AGE
observability-app-5c4d8dc8c-gt4h4	0/1	InvalidImageName	0	16h
observability-app-654cccc487-pgch9	1/1	Running	0	23h
observability-app-654cccc487-sqshp	1/1	Running	1 (12s ago)	23h

```
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> kubectl get deployment observability-app -n observability-demo
```

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
observability-app	2/2	1	2	23h

```
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> _
```

Second pod restart proof with restart count.

18. Azure DevOps Pipeline YAML

Section goal

Build and push image automatically using a self-hosted agent.

```
trigger:
- main

pool:
  name: Default

variables:
  imageRepository: &#x27;observability-app&#x27;
  dockerfilePath: &#x27;Dockerfile&#x27;
  tag: &#x27;$(Build.BuildId)&#x27;

steps:
- task: Docker@2
  displayName: Build and push Docker image to Azure Container Registry
  inputs:
    command: buildAndPush
    repository: $(imageRepository)
    dockerfile: $(dockerfilePath)
    containerRegistry: &#x27;acr-observability-service-connection&#x27;
    tags: |
      $(tag)
      latest
```

```
PS C:\Users\Admin\Documents\azure-monitor-aks-observability-capstone> git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   azure-pipelines.yml

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        screenshots/08-azure-account-verified.png
        screenshots/09-resource-group-created.png
        screenshots/10-acr-created.png
        screenshots/11-image-built-and-pushed-to-acr.png
        screenshots/12-acr-image-tags.png
        screenshots/13-log-analytics-workspace-created.png
        screenshots/14-kubectl-get-nodes.png
        screenshots/15-aks-addons-verified.png
        screenshots/16-kubernetes-files-applied.png
        screenshots/17-kubernetes-resources-running.png
        screenshots/18-app-working-through-port-forward.png
        screenshots/19-kubectl-get-nodes.png
        screenshots/20-aks-addons-verified.png
        screenshots/21-kubernetes-files-applied.png
        screenshots/22-ingress-created.png
        screenshots/23-app-endpoints-tested-through-appgw.png
        screenshots/24-aks-application-logs.png
        screenshots/25-aks-pods-wide-output.png

no changes added to commit (use "git add" and/or "git commit -a")
```

Pipeline YAML added and committed.

19. Azure DevOps Service Connection and Self-Hosted Agent

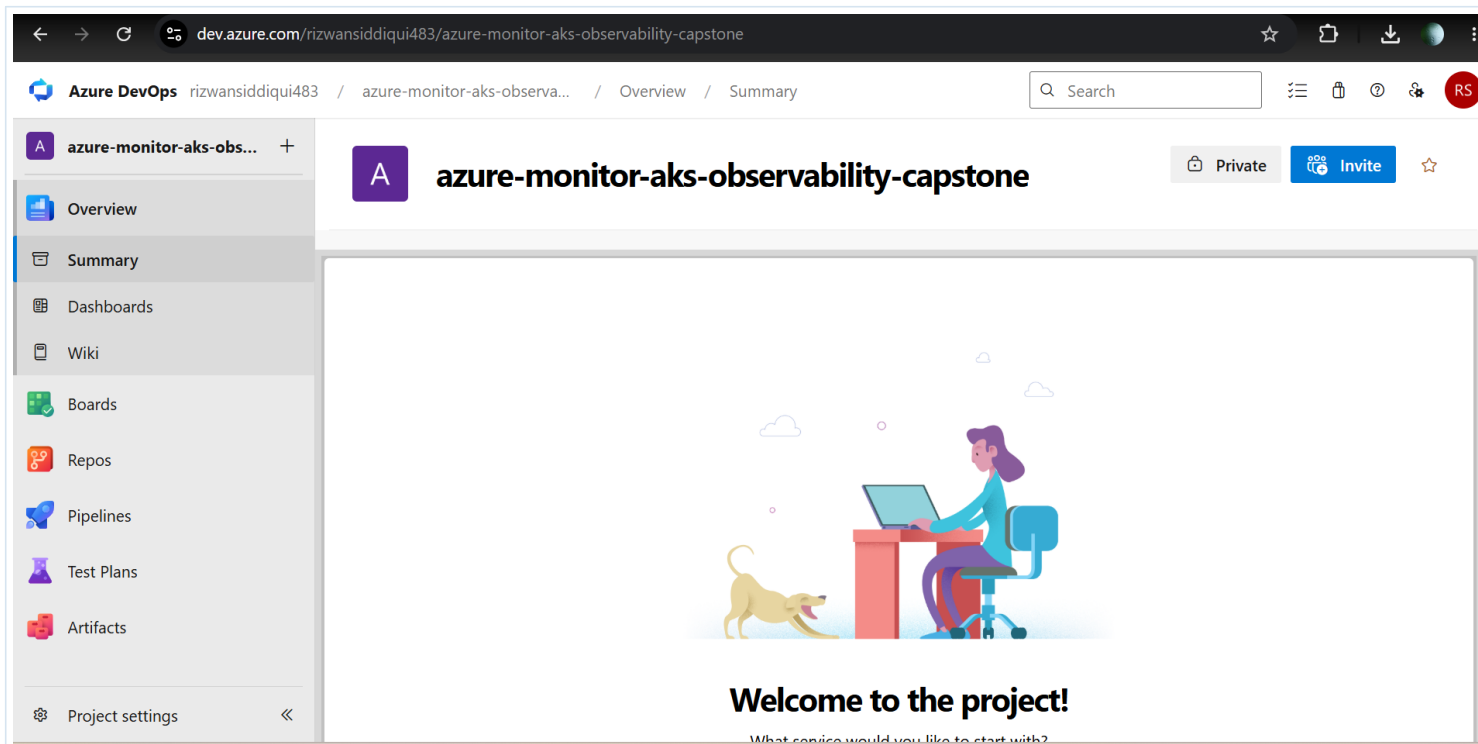
Section goal

Support pipeline execution when Microsoft-hosted parallelism is not available.

```
# Create or update ACR credentials only for lab validation
az acr update --name acrriizwanobs268 --admin-enabled true
az acr credential show --name acrriizwanobs268
# Do not screenshot the password output.

# Configure agent under C:\azagent after downloading agent zip
cd C:\azagent
.\config.cmd
# Server URL: https://dev.azure.com/rizwansiddiqui483
# Authentication: PAT
# Pool: Default
# Agent name: Rizwan-Local-Agent
# Run as service: N

.\run.cmd
```

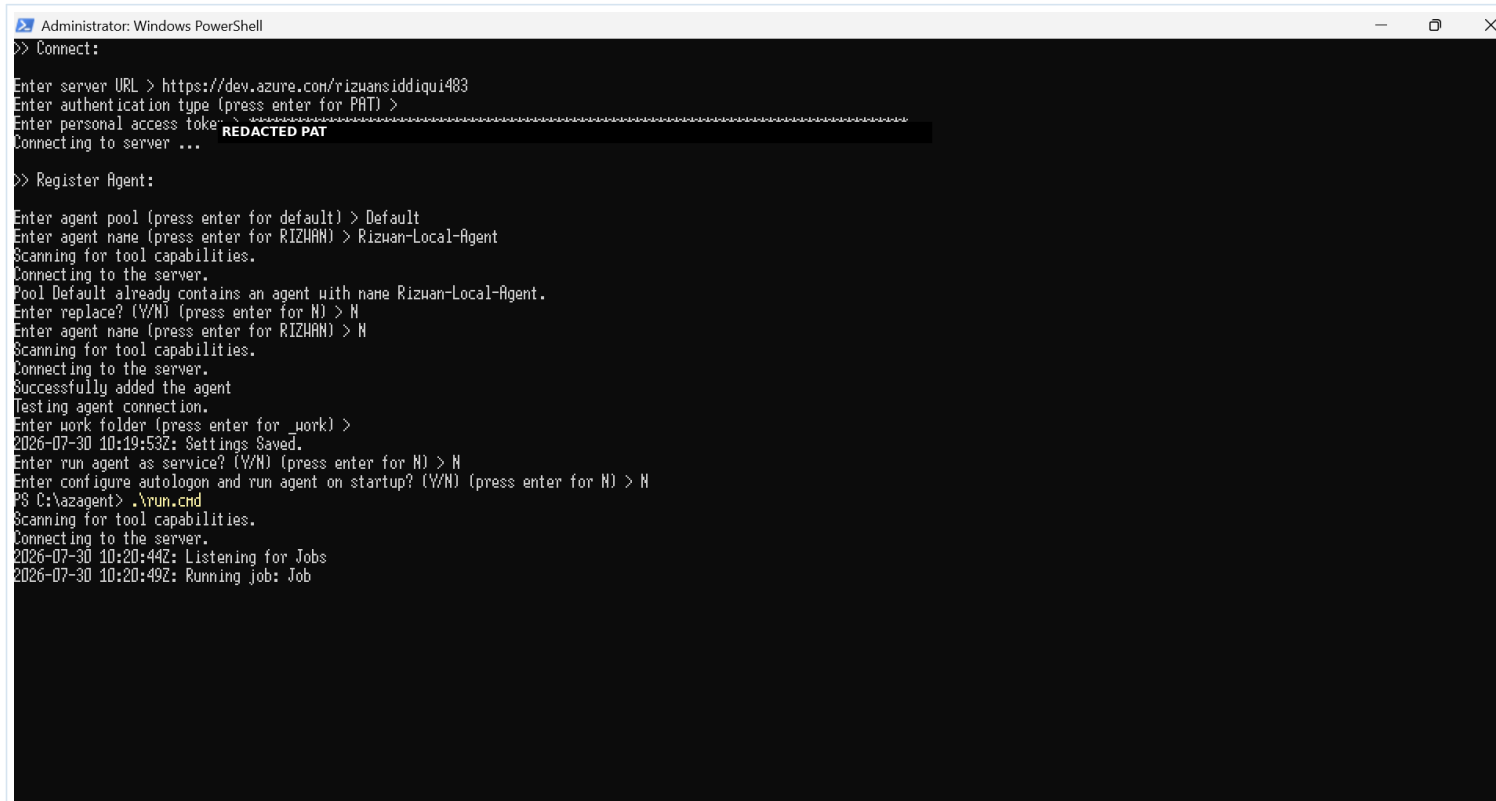


Azure DevOps project created.

The screenshot displays the Azure DevOps web interface. The browser address bar shows the URL: `dev.azure.com/rizwansiddiqui483/azure-monitor-aks-observability-capstone/_settings/adminservices`. The breadcrumb navigation indicates the path: `rizwansiddiqui483 / azure-monitor-aks-observability-capstone / Settings / Service connections`. The left-hand navigation pane is expanded to 'Project Settings', showing a list of settings categories including General, Overview, Teams, Permissions, Notifications, Service hooks, Dashboards, Boards, Project configuration, Team configuration, GitHub connections, and Pipelines. The main content area is titled 'Service connections' and features a 'Security' tab and a 'New service connection' button. Below this, there is a filter bar with the text 'Filter by keywords' and dropdown menus for 'Created by', 'Type', and 'Authentication method'. A table lists the existing service connections:

Name	Type	Authentication method
acr-observability-service-connection	Docker Registry	Basic Authentication

ACR Docker Registry service connection created.



```
Administrator: Windows PowerShell
>> Connect:

Enter server URL > https://dev.azure.com/rizwansiddiqui483
Enter authentication type (press enter for PAT) >
Enter personal access token > REDACTED PAT
Connecting to server ...

>> Register Agent:

Enter agent pool (press enter for default) > Default
Enter agent name (press enter for RIZWAN) > Rizwan-Local-Agent
Scanning for tool capabilities.
Connecting to the server.
Pool Default already contains an agent with name Rizwan-Local-Agent.
Enter replace? (Y/N) (press enter for N) > N
Enter agent name (press enter for RIZWAN) > N
Scanning for tool capabilities.
Connecting to the server.
Successfully added the agent
Testing agent connection.
Enter work folder (press enter for _work) >
2026-07-30 10:19:53Z: Settings Saved.
Enter run agent as service? (Y/N) (press enter for N) > N
Enter configure autologon and run agent on startup? (Y/N) (press enter for N) > N
PS C:\azagent> .\run.cmd
Scanning for tool capabilities.
Connecting to the server.
2026-07-30 10:20:44Z: Listening for Jobs
2026-07-30 10:20:49Z: Running job: Job
```

Self-hosted agent listening and running a job. PAT line is masked.

20. Pipeline Troubleshooting and Success

Section goal

Fix daemon and authentication failures, then verify tags.

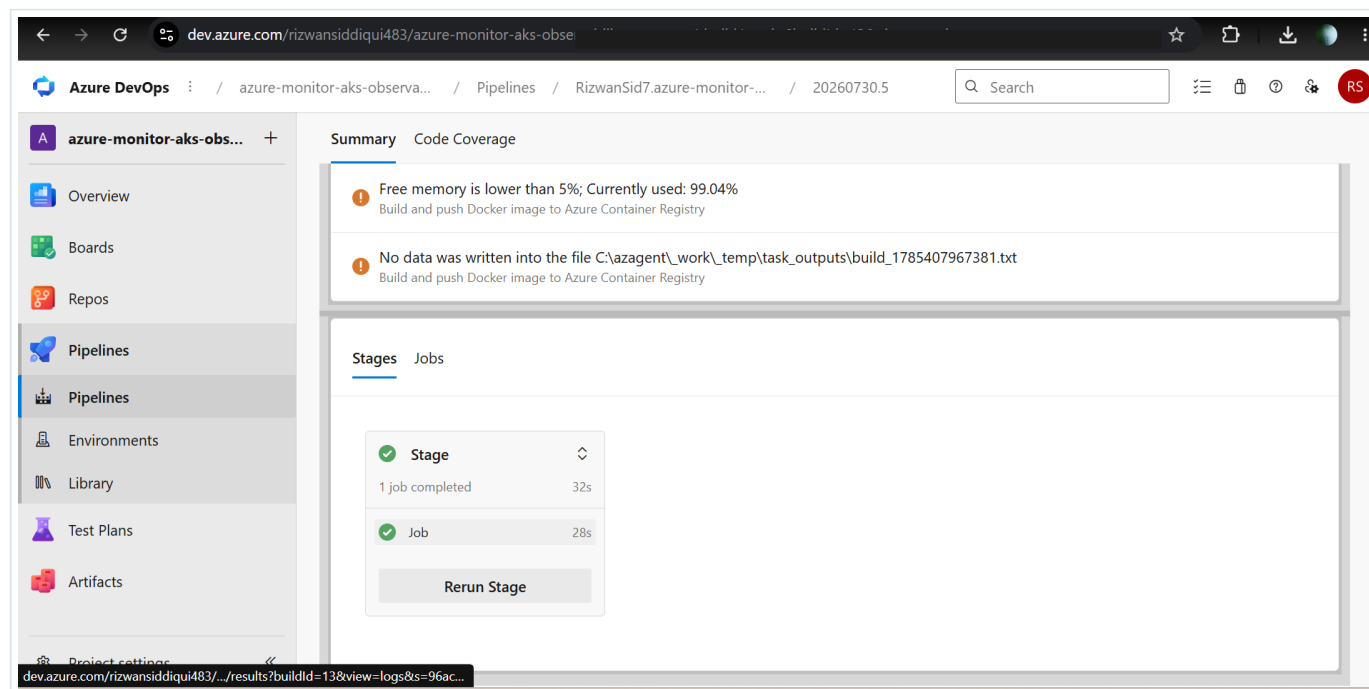
Failure	Root cause	Fix
Pipeline queued	Self-hosted agent not running in Default pool.	Run C:\azagent\run.cmd and keep window open.
Docker daemon error	Docker Desktop not running on agent machine.	Start Docker Desktop, verify docker ps, rerun pipeline.
ACR authentication required	Service connection credentials invalid or admin disabled before validation.	Refresh ACR credential/service connection and rerun. Disable admin after success.
Low memory warning	Local machine memory almost full.	Close apps, free memory, rerun if build fails.

The screenshot shows the Azure DevOps web interface. On the left, the 'Pipelines' tab is selected in the sidebar. The main view shows 'Jobs in run #20260...' for the pipeline 'RizwanSid7.azure-monitor-aks-observability-capstone'. The job list includes:

- Initialize job: Success (<1s)
- Checkout Rizwan...: Success (5s)
- Build and push D...: Failure (1s)
- Post-job: Check...: Success (<1s)
- Finalize Job: Success (<1s)

The 'Build and push D...' job is expanded, showing a detailed error log. The log indicates a failure to connect to the Docker API at 'npipe:////./pipe/docker_engine' and mentions 'ERROR: failed to connect to the docker API'.

Pipeline failed because Docker daemon was not running.



Pipeline succeeded after Docker and ACR auth fixes.

```
}
PS C:\WINDOWS\system32> az acr repository show-tags --name acrrizwanobs268 --repository observability-app --output table
Result
-----
13
latest
PS C:\WINDOWS\system32>
```

ACR tags after pipeline: build ID and latest.

21. Git Commands for Final Documentation and Screenshots

Section goal

Commit README, docs, and redacted screenshots safely.

```
# Final documentation
git status
git add README.md docs/ monitoring/ k8s/ azure-pipelines.yml
git commit -m "Added final README architecture and demo documentation"
git push origin main

# Replace screenshots with redacted versions
git add -u screenshots/
git add screenshots/*.png
git commit -m "Replaced screenshots with final redacted versions"
git push origin main

# Verify clean state
git status
git log --oneline --max-count=5
```

Screenshot safety checklist

Before pushing, verify that PAT token, ACR password, account email, subscription ID, tenant ID, client ID, object ID, workspace ID, full resource IDs, and public/private IPs are masked.

22. Production Improvements

Section goal

Make the lab production-ready before real customer use.

Production Readiness and Best Practices

Area	Recommendation
Networking	Use private AKS where possible, private endpoints for ACR, restricted inbound traffic, and WAF when exposing internet workloads.
Identity	Use managed identity, least-privilege RBAC, separate service connections, and avoid long-lived admin passwords.
Delivery	Use protected branches, pull requests, pipeline approvals, environment gates, and image scanning before deployment.
Reliability	Use multiple node pools, autoscaler, readiness/liveness probes, rolling updates, PodDisruptionBudgets, and zones where available.
Observability	Use Container Insights, Prometheus/Grafana where needed, KQL queries, alert rules, dashboards, and clear incident runbooks.
Cost	Use budgets and alerts, right-size nodes, set log retention, stop/delete lab resources, and avoid leaving Application Gateway and AKS nodes running unnecessarily.
Security	Use HTTPS/TLS, Key Vault, Defender for Cloud, network policies, secret management, and restrict public IP exposure.

23. Cleanup Commands

Section goal

Delete lab resources after recording the demo video.

Important

Do not delete Azure resources until the final demo video, portal screenshots, pipeline proof, and ACR tag proof are captured.

```
$RG="rg-aks-observability-capstone";

# Delete entire lab resource group
az group delete --name $RG --yes --no-wait

# Verify deletion after a few minutes
az group exists --name $RG
# Expected: false

# Save screenshot as: 51-azure-resources-deleted-proof.png
git add screenshots/51-azure-resources-deleted-proof.png
git commit -m "Added Azure resource cleanup proof"
git push origin main
```

Official Microsoft Documentation References

Use these primary Microsoft sources whenever you need to verify steps, product behavior, limits, and pricing. Pricing pages are estimates and should be checked again before production decisions.

Topic	Official link
AKS monitoring and Container Insights	https://learn.microsoft.com/en-us/azure/aks/monitor-aks
Application Gateway Ingress Controller overview	https://learn.microsoft.com/en-us/azure/application-gateway/ingress-controller-overview
AGIC annotations	https://learn.microsoft.com/en-us/azure/application-gateway/ingress-controller-annotations
Create AGIC add-on for a new AKS cluster	https://learn.microsoft.com/en-us/azure/application-gateway/tutorial-ingress-controller-add-on-new
Azure Container Registry documentation	https://learn.microsoft.com/en-us/azure/container-registry/
ACR quickstart with Azure CLI	https://learn.microsoft.com/en-us/azure/container-registry/container-registry-get-started-azure-cli
ACR Tasks: build and run container images	https://learn.microsoft.com/en-us/azure/container-registry/container-registry-quickstart-task-cli
Azure Monitor Log Analytics overview	https://learn.microsoft.com/en-us/azure/azure-monitor/logs/log-analytics-overview
Log queries in Azure Monitor	https://learn.microsoft.com/en-us/azure/azure-monitor/logs/log-query-overview
Azure CLI Log Analytics query	https://learn.microsoft.com/en-us/azure/azure-monitor/logs/api/request-format
Azure DevOps self-hosted agents	https://learn.microsoft.com/en-us/azure/devops/pipelines/agents/docker?view=azure-devops
Docker task for Azure Pipelines	https://learn.microsoft.com/en-us/azure/devops/pipelines/tasks/reference/docker-v0?view=azure-pipelines
AKS best practices	https://learn.microsoft.com/en-us/azure/aks/best-practices
AKS baseline architecture	https://learn.microsoft.com/en-gb/azure/architecture/reference-architectures/containers/aks/baseline-aks
AKS pricing	https://azure.microsoft.com/en-us/pricing/details/kubernetes-service/
Application Gateway pricing	https://azure.microsoft.com/en-in/pricing/details/application-gateway/
Azure Monitor pricing	https://azure.microsoft.com/en-in/pricing/details/monitor/
Azure Container Registry pricing	https://azure.microsoft.com/en-gb/pricing/details/container-registry/